Abu Dhabi City Campus Daily/Weekly Planner

Week 12: 16th – 20th November 2026

Subject: Mathematics **Unit Title: Calculus** **Grade: 12**

Lessons	Learning targets	Assessment tools and strategies Formative/ Summative/ Differentiation	Resources and links for Parents
Lesson 1: Product Rule	I can apply the product rule to differentiate functions involving the product of two expressions.	2.1 Clarify the meaning of factors, products and derivative notation using colour-coded expressions and worked examples.	**Related Videos / Resources** Derivatives of Trig Functions | Calculus Lesson 12 Derivatives of Trig Functions Examples | Calculus The Product Rule | Calculus Lesson 13 Product Rule Examples | Calculus The Quotient Rule | Calculus Lesson 14 Quotient Rule Examples | Calculus
Lesson 2: Quotient Rule	I can apply the quotient rule to differentiate functions involving the division of two expressions.	2.3 Support decoding of the quotient-rule structure with colour-coded numerator and denominator functions.	
Lesson 3: Derivatives of Trigonometric Functions	I can differentiate basic sine and cosine functions and apply the appropriate derivative rules to simple trigonometric expressions.	5.3 Build fluency through modelled examples and graduated practice with constant multiples and sums.	
Lesson 4: Formative Assessment 2 – Unit 2	I can apply different differentiation techniques, including the product, quotient and trigonometric derivative rules, to solve mathematical problems.	9.3 Develop self-assessment through checking solutions, identifying errors and reflecting on next steps.	
Homework	Product Rule, Quotient Rule and Derivative of Trigonometric Functions worksheets on Teams assigned as homework.		
Teacher Notes	Students should bring a GDC/graphing calculator, notebook and device. Ensure access to lesson PowerPoints, Teams/OneNote resources and practice materials. Review the product and quotient rule structures before the formative assessment.		

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Technology Integration	Microsoft Teams; OneNote; GDC/graphing calculator; Desmos; lesson PowerPoints; digital practice and assessment materials.
Cross-curricular links	Physics/Science: Apply product, quotient and trigonometric derivatives to model changing quantities, motion and periodic phenomena. Engineering: Interpret rates of change in systems involving multiple variables and functions. Technology: Use graphing tools to compare functions with their derivatives.
External Assessment Integration	Direct alignment with AP Calculus AB Unit 2: applying differentiation rules, including the product rule, quotient rule and derivatives of sine and cosine. Include AP-style multiple-choice and free-response questions requiring accurate notation, algebraic manipulation and interpretation.
Wellbeing/ ATL Skills	Thinking Skills: Select and apply appropriate differentiation rules; connect algebraic and graphical representations; analyse errors and verify results. Communication Skills: Use correct derivative notation and explain each step in a clear mathematical sequence. Self-Management Skills: Organise multi-step solutions, manage assessment time and use feedback to identify areas for improvement.

Standards	AP Calculus AB Unit 2: Product Rule; Quotient Rule Derivatives of sine and cosine functions Application of differentiation rules to algebraic and trigonometric expressions.
Inquiry/ TOK Question	How can the structure of a function determine the differentiation rule we choose, and how can its derivative help us interpret how the function changes?
Success Criteria (Measurable)	**Lesson 1: Product Rule** • **All students must** identify the two factors and apply the product rule correctly to differentiate a simple product of functions. • **Most students should** differentiate products involving polynomial expressions and simplify the resulting derivative accurately. • **Some students could** solve unfamiliar product-rule problems and explain why differentiating each factor separately is not sufficient. **Lesson 2: Quotient Rule** • **All students must** identify the numerator and denominator and use the quotient rule with correct order and notation. • **Most students should** differentiate quotients of polynomial functions and simplify the resulting expression correctly. • **Some students could** solve complex quotient-rule problems, compare alternative methods and justify the efficiency of their approach. **Lesson 3: Derivatives of Trigonometric Functions** • **All students must** recall and apply the derivatives of sine and cosine functions. • **Most students should** differentiate simple trigonometric expressions involving constant multiples and sums or differences. • **Some students could** combine trigonometric derivatives with other differentiation rules and interpret the derivative in context. **Lesson 4: Formative Assessment –Unit 2** • **All students must** demonstrate the correct use of product, quotient and basic trigonometric derivative rules in routine questions. • **Most students should** select and apply the appropriate differentiation technique to multi-step expressions and show clear working. • **Some students could** solve unfamiliar, higher-order problems involving multiple differentiation rules and justify their mathematical reasoning.
National Identity Domain	**Respect**
	Area: Citizenship
	Implementation: Encourage perseverance, academic integrity and reflection as students respond to challenging differentiation problems and use assessment feedback to improve.